

DECENTRALIZED CONTROL DESIGN: AN OVERVIEW

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ABSTRACT. In this paper we give an overview of decentralized decision making and control research and try to discern its trends. We first examine the decentralized structures which arise in a number of application areas. We then consider the stabilization of composite systems which are subjected to decentralized information constraints and we also examine appropriate design techniques for such systems. This is followed by a presentation of the basic results on stochastic decentralized and partially decentralized control problems where non-classical information patterns arise. Decentralized state estimation is also briefly discussed and the robustness issues in decentralized control are addressed. Some conclusions are then given on promising future research directions.

INTRODUCTION

The decentralized control problem arises when a system is acted upon by a number of different controllers which are not able to communicate amongst themselves on-line or are not in a position to process on-line the vast amount of sensor data available to them. The controllers may be physical devices or they may be human decision makers. It is however assumed that the decision making is cooperative i. e. that the controllers may use a priori agreed rules to process the on-line information in order to achieve some overall system objective. This kind of a situation is very different from the one where the decentralized decision makers have individual goals which are partially or wholly in conflict with those of the overall system and where we need to use game theoretic ideas (like Nash equilibrium) to do the analysis. Such conflict situations will not be dealt with in this paper.

A significant body of knowledge has evolved in the last 15 years on the analysis and design of decentralized control systems. In this paper, we will classify this knowledge and highlight some of the more interesting results which have been obtained.

It should be emphasized that although decentralized controllers have been designed and used for controlling interconnected dynamical systems for

over two decades, the design was based on ad hoc methods. This is still the case for most industrial and management systems design. The reasons for this are briefly discussed in the next section where we examine some applications and discuss the most important issues associated with the design of decentralized control systems.

With the increased interest in multi-variable control systems and in optimization techniques there have been attempts to translate the basic results concerning centralized controller design to decentralized situations. From these studies, numerous difficulties associated with decentralized control became apparent. For example, one of the best known and most useful results in the theory of control for centralized systems is the certainty equivalence property. This, broadly speaking, states that for linear dynamical systems acted upon by white noise disturbances, it is possible to design a controller which minimises the expected value of a quadratic cost function by designing separately an optimal controller (by solving a deterministic design problem) and an optimal state estimator. This result fails in the case of decentralized control due to the non-classical information pattern which arises when we have a decentralized information structure. Similarly, one of the most important results concerning stabilisation of linear time in-

variant systems, which states that by applying an appropriate time invariant linear dynamical control law, one can place the poles of the system arbitrarily if and only if this system is both controllable and observable, also fails under a decentralized information structure.

The above issues have been investigated in depth and numerous interesting results are available. These will be discussed later in this paper (in sections 3, 4 and 5).

Recently considerable interest has been shown in defining and investigating sensitivity and robustness properties of control systems. This issue is related to the classical concepts of gain and phase margins, and consists of providing for a satisfactory operation of a control system under some class of perturbations in plant behaviour, in informational links or in the controller itself. For example, it was shown that with an appropriate choice of a performance index, linear quadratic design methods have desirable sensitivity and robustness properties according to various criteria (Safonov and Athans [1977]). Obviously the robustness issues are of importance in the case of decentralized control structures. This aspect will be briefly discussed later in this paper.

Finally, it should be noted that the design of decentralized control structures is closely associated with the notions of process decomposition (partitioning), large scale system stability (stability tests based upon aggregated data) and model reduction techniques. These issues have received a lot of attention and a significant number of results are available. In order to concentrate on some key topics in decentralized control we will not discuss the above notions in this paper.

We would like to point out that despite all the efforts put into the development of the theory one still feels that the fundamental conclusion of Sandell et al [1978] as formulated about 5 years ago is still valid. Namely, we believe that there is a very long way to go before a unified theory of decentralized control emerges - including, besides the traditional stability and performance index issues, also the costs of communications, reliability aspects, the costs of hardware and software, the value of information (or the lack of information), and, eventually, the features due to human factors being present in certain important applications.

IMPORTANT FEATURES OF DECENTRALIZED CONTROL APPLICATIONS

In the real world decentralized control situations arise in many different fields. In engineering, typical examples would be:

- 1) Power systems, where different parts of the network and different power stations are under the control of different authorities. These authorities (area controllers) would have appropriate a priori specific rules for acting, based upon previous knowledge and plans for local distribution but they do not have either the time for exchanging all the information on-line nor the physical possibility to perform very complicated computations on-line.
- 2) Process control systems where for large plants operating in the steady state the local follow-up controllers need to regulate the controlled variables to their desired set-points, to stabilise the process and to reduce the impact of fast disturbances acting upon the plant. To achieve the centralisation of an on-line data base, may be either physically impossible (due to time limitations) or very costly. Even in the case when it is possible to gather all on-line information in one place, the decentralized control structure may be (and usually is) preferred due to its decreased on-line computing requirements and because of the possibility of a better handling of emergency situations. For example, if a single-input-single-output control loop fails then the process operator would usually know what to do in order to take corrective action. If, however, something goes wrong with the multivariable controller, then even the experienced operator may have great difficulty in deciding on what corrective action to take.
- 3) Fast complex servomechanisms e.g. industrial robots where the movement (trajectory follow-up) control requires immediate reaction to incoming measurement data. Here there is not enough time to calculate each control input value using the complete data set.
- 4) Telephone traffic routing (Singh [1983]) where the decisions related to current operation of traffic transit centres have to be made individually in each centre so as to adjust the load sharing parameters in order to distribute the traffic according to the pattern as specified at the level of supervision.
- 5) Flood control in a multireservoir

system where the individual reservoir operators may have decision rules (periodically adjustable perhaps by the central dispatcher) by which they specify the water releases. Their decisions are based upon on-line local information for several reasons. One of these is to provide for a satisfactory operation even if all information links between the operators are cut. Also the operators should be given the control rules which they can accept and understand. This makes it even harder to design the rules with structural properties which provide for a satisfactory operation of the system as a whole.

Without going into details specific for the above applications, we can roughly classify the decentralized control problems as follows.

A. Follow-up control situation, where a desired system behaviour e. g. state trajectory has been determined by a higher layer (Findeisen et al [1980]) control mechanism operating on a slower time scale and where the local controllers aim to keep the system operating as desired. Here we can distinguish between essentially two cases:

A1. The system is to be operated at steady state over a long period of time (this is typical in process control, and at the lower levels of telephone routing networks (Bernussou and Titli [1982])). Then in order to study the behaviour of this system and to design appropriate decentralized follow-up control rules, we may use a linear time invariant description of the systems dynamics valid locally "near" the operating point. Also the performance measure can be selected for example in the form of a quadratic performance index and the posing of constraints on control inputs and state variables can be avoided because of the "small" anticipated deviations from the steady state values.

A2. The reference trajectory can be rapidly changing and it may be unknown in advance (typical case for complex servomechanisms). Then the design of the decentralized controllers need to be based upon a nonlinear system model and it should take into account saturation effects (in actuators), constraints on state variables etc.

B. "Management type" control situation, where the local controllers have to exercise appropriate control actions upon a system in order to provide for a satisfactory behaviour of this system. There is not, however, a refe-

rence operating point or a reference trajectory. At the same time the control horizon is usually finite. The flood control example is a good illustration of such a situation.

The available theoretical results associated with decentralized control problems concern mostly the case A1 since linear time invariant dynamical systems can be investigated relatively efficiently from a qualitative point of view. Also some quantitative analysis is possible, although it is surprising how little can be done in terms of quantitative analysis even for linear systems. The case A2 is far more complicated and there is little more available so far than some stability analysis of decentralized control schemes for nonlinear systems. The case B is even more complex and difficult to examine.

The above discussion explains why it is still the case that practical design is based upon ad-hoc procedures and upon a physical understanding of the properties and real-life requirements associated with a particular case study (like in the flood control case). In fact, it would appear that it might be more appropriate to develop first a methodology for the design, say, of decentralized control structures for multireservoir systems, or decentralized controllers for chemical plants etc. instead of trying to build up a unifying theory covering different fields of applications since existing theory is not completely suited to the tackling of such practical problems. Nevertheless, a lot of progress has been made in the understanding of decentralized control systems. One convenient way of classifying the key elements of this knowledge is to consider first the split between deterministic decentralized control problems and stochastic decentralized control problems (or in general decentralized control under uncertainty). In the former case the key notion in the analysis and controller design arises from the stability issue and concerns the coordinability properties under decentralized information structures. In particular, the presence or absence of decentralized "fixed modes" is important, and it is necessary to get around their effects. Performance-index-based design techniques have been developed for fixed structure controllers.

In the case of stochastic decentralized control, the main question arises with respect to the structural properties of the optimal control schemes which result when information constraints are imposed. These cons-

straints do not allow for optimal solutions with separation between state estimation and the design of control laws. By enforcing this separation we can consider fixed structure linear controllers as a practical way of designing the controls within a stochastic control framework. Finally, when we use a deterministic problem formulation in order to design control laws then before this decentralized control can be implemented we often need to do decentralized observation or filtering. This we will briefly discuss later in the paper.

DECENTRALIZED STABILIZATION OF LINEAR TIME-INVARIANT SYSTEM

In this section we consider the class of linear systems described by

$$\dot{x}(t) = Ax(t) + \sum_{i=1}^v B_i u_i(t) \quad (1a)$$

$$y_i(t) = C_i x(t) \quad (1b)$$

$$x \in R^n, y_i \in R^{r_i}, u_i \in R^{m_i}$$

where v , is the number of control stations. At each control station only the local outputs $y_i(t)$ are observed and the local control input $u_i(t)$ needs to be specified. Thus (1b) defines a decentralized information structure and we need to study the controllability of system (1) under such information structures. This controllability can be defined as follows [Kobayashi et al, 1978; Kobayashi and Yoshikawa, 1982]; the system (1) is controllable under the decentralized information structure if there exists a decentralized control law $u_i(t) = \phi_i(I_i(t))$, $i=1, \dots, v$, which transfers any initial state, which is not initially known to the control stations, to the origin in a finite interval of time. $I_i(t)$ denotes the set of data available for control station i at time t ; $I_i(t) = \{y_i(\tau), u_i(s); \tau \in [0, t], s \in [0, t)\}$.

It is important to understand that in decentralized control systems, a control station can transmit information to other control stations through the state space by using signalling strategies [Sandell and Athans, 1975]. As a result of this, the system which is controllable under the decentralized information structure may not even be stabilizable by the following linear time-invariant controllers

$$\dot{z}_i(t) = S_i z_i(t) + T_i y_i(t) \quad (2a)$$

$$u_i(t) = L_i z_i(t) + K_i y_i(t) + V_i(t), \quad i=1, \dots, v \quad (2b)$$

where $z_i(t) \in R^{n_{oi}}$ and $V_i(t) \in R^{m_i}$ is an external input. The reason for this is that the signalling requires time-varying and/or nonlinear control strategies to be applied. The pole placement and stabilization under the decentralized control structure, as given by (2), is related to the notion of the fixed modes.

Definition [Wang and Davison, 1973]

Let $\mathbb{K} = \{K | K = \text{block-diagonal } [K_1, \dots, K_v],$

$$K_i \in R^{m_i \times r_i}, i=1, \dots, v\}, \quad (3)$$

Let $B = [B_1, \dots, B_v]$ and $C = [C_1^T, \dots, C_v^T]^T$, where B_i, C_j are specified by (1).

Then the set of fixed modes of system (1) for a decentralized control (with respect to set $\mathbb{K}$ of block diagonal output feedback matrices) is defined by:

$$\Lambda(C, A, B, \mathbb{K}) \stackrel{\text{df}}{=} \bigcap_{K \in \mathbb{K}} \sigma(A + BKC) \quad (4)$$

where $\sigma(x) =$ the set of eigenvalues of x .

Thus the fixed modes of the system (1) are those poles of the open-loop system, which cannot be moved by applying a decentralized static time-invariant output feedback control law.

The fundamental result of Wang and Davison [1973] is:

Theorem 1

(i) The closed-loop system as given by eqs. (1), (2), is stabilizable by an appropriate choice of matrices S_i, T_i, L_i, K_i , $i=1, \dots, v$ if and only if $\Lambda(C, A, B, \mathbb{K}) \subset C^-$, where C^- is the complex left plane and $\mathbb{K}$ is the set of block diagonal output feedback matrices as given by (3).

(ii) All the poles of the closed-loop system (1), (2) can be placed arbitrarily (subject to the conjugate-pair condition) by an appropriate choice of the matrices S_i, T_i, L_i, K_i , $i=1, \dots, v$ if and only if $\Lambda(C, A, B, \mathbb{K}) = \emptyset$. The design procedures for choosing matrices S_i, T_i, L_i, K_i are readily available. In view of the above result it is not surprising that a lot of research work has been devoted recently to provide for the characterisation of the fixed modes and to find ways of getting around their effects if there are unstable fixed modes.

Let us therefore consider in some detail the characterisation of the fixed modes and controllability under decentralized information structures. First of all it is obvious that $\Lambda_{\text{cent}} \subset \Lambda(C, A, B, K)$ where Λ_{cent} is the set of unobservable and uncontrollable modes of system (1) under a centralized information structure. If there are uncontrollable or unobservable (centrally) modes which are unstable, then... well, one needs to change the system itself! Thus, in what follows we will assume that the triplet (C, A, B) is controllable and observable and the fixed modes will arise only due to the decentralization.

First of all, the fixed modes can be computed (see Definition (4)) by computing first the eigenvalues of A and then the eigenvalues of $A + BKC$ for randomly chosen block-diagonal matrix K . The common elements of these two sets are then (with Probability 1) the fixed modes of system (1). The calculation of the eigenvalues should not cause any essential difficulties even for high dimensional systems although there may be problems with roundoff errors. This test does not give us information on the nature (origin) of any possible fixed modes.

Algebraic characterisation of fixed modes was provided by Anderson and Clements [1981]. Their result is:

Theorem 2

λ^0 is a fixed mode of system (1) for a decentralized control if and only if there exist disjoint sets of indices $\alpha = \{i_1, \dots, i_k\}$, $\beta = \{i_{k+1}, \dots, i_v\}$, $\alpha \cup \beta = \{1, \dots, v\}$ such that

$$\text{rank} \begin{bmatrix} \lambda^0 I - A & B_\alpha \\ C_\beta & 0 \end{bmatrix} < n \quad (5)$$

where $B = [B_{i_1}, \dots, B_{i_k}]$, $C = [C_{i_{k+1}}, \dots, C_{i_v}]^T$.

It is obvious that condition (5) needs to be checked only for the eigenvalues of A .

The above result has an immediate useful interpretation: if λ^0 is a fixed mode of (1), then for some disjoint partition of information (1a) between two "aggregate" control stations λ^0 is simultaneously uncontrollable from one of these stations and unobservable from the other station (see Figure 1).

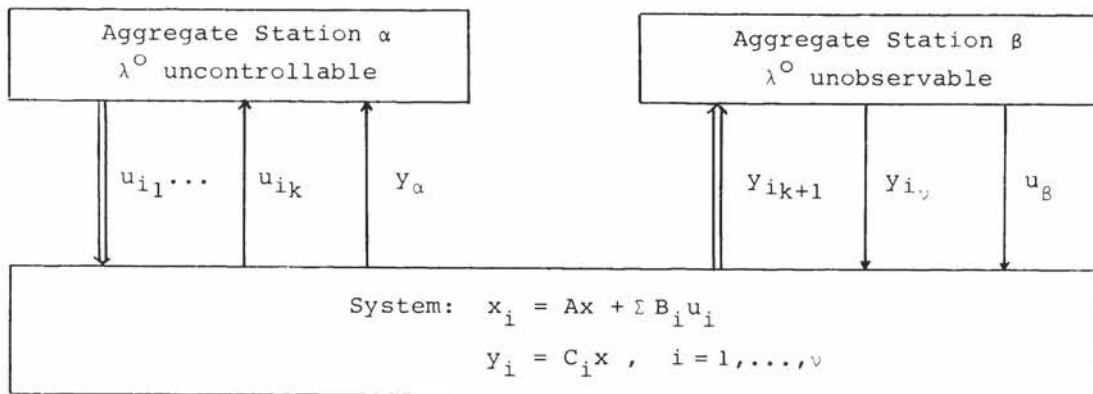


Fig. 1. Characterisation of a fixed mode λ^0 .

A useful conclusion from the above is that if $\lambda \in \sigma(A)$ is controllable and observable w.r.t. (A, B_i) and (C_i, A) for some $i \in \{1, \dots, v\}$ then λ cannot be a fixed mode of (1). Anderson and Clements [1981] provided also for a characterisation of fixed modes by using a transfer function matrix description of system (1). This result was further refined by Anderson [1982]. He showed also that it was possible to define and to characterise the degree of a fixed mode. Seraji [1982] provided another form of a fixed mode characterisation using also a transfer function matrix description. It

is beyond the scope of this paper to present these interesting results in more detail. The reader is referred to a recent paper by Davison and Ozguner [1983] for extensive characterisations of decentralized fixed modes.

Let us now turn back to the issue of controllability under the decentralized information structure. This was investigated in detail by Kobayashi and Yoshikawa [1982] who made use of the approach by Corfmat and Morse [1976]. The most interesting result of Kobayashi [1982] concerns the equivalence between controllability under the decentralized information

structure and the nonexistence of fixed modes of the so-called quotient system. Without going into detail, we can use the system as shown in Fig.1 with two aggregated control stations α and β to explain the basic idea. Let λ^0 be a fixed mode of system (1) as characterised in Fig.1. Then, assuming that the triplet (C, A, B) is both controllable and observable, λ^0 must be observable from station α (through y_α) and controllable from station β (through u_β). It appears that if

$$C_\beta(sI - A)^{-1}B_\alpha \neq 0 \quad (6)$$

then by using an appropriate strategy (Kobayashi and Yoshikawa [1982]), station α can transmit through the state space sufficient information to control the mode associated with λ^0 . Thus under condition (6) λ^0 is a fixed mode of (1) but is controllable under the decentralized information structure. However, if condition (6) is not satisfied then λ^0 is uncontrollable under the decentralized information structure - in terms of Kobayashi and Yoshikawa, [1982] λ^0 is a fixed mode of the quotient system.

From a practical point of view, it is important to know whether the existence of a given fixed mode is due to structural properties of the system (1) or whether it arises because of particular values of nonzero system parameters. In general, it appears that (especially in large systems) many entries in the matrices A, B, C have fixed zero values associated with the structural properties of the system. The values of the remaining parameters result from system identification. Let us denote the space of these "nonzero" parameters by R^q . Now, if for every vector $p \in R^q$ of nonzero entries of A, B, C λ is a fixed mode of system (1) then λ is termed as a structurally fixed mode, Sezer and Siljak [1981]. On the other, if for some vector of parameters $p \in R^q$ a given $\lambda \in \sigma(A)$ is not a fixed mode of (1) then λ is not a fixed mode for almost all $p \in R^q$.

Referring to partition of system inputs and outputs between the two imaginary "aggregate" stations as given in fig.1, we can clarify the structurally fixed modes into two types. The diagram presented in fig.2 is related to the situation in which the state vector can be split into three sub-vectors i.e. $x = (x^1, x^2, x^3)$ such that there is no direct or indirect access from the group of inputs u_α to x^2 and x^3 . Further, neither x^1 nor x^2 can be observed through y_β and no

signalling from station α to station β through the state space is possible due to the block upper triangular form of the matrix A . These properties are of course of a structural nature. The modes associated with the second group of state variables x^2 are structurally fixed modes of type 1. Obviously they are also uncontrollable under the decentralized information structure. The structurally fixed mode (or modes) of type 2 exists if for some two disjoint sets of indices α, β as defined in theorem 2 the following condition is satisfied, Seraji [1982]:

$$\bar{\delta} \begin{bmatrix} A & B \\ C_\beta & 0 \end{bmatrix} < n, \quad (7)$$

where $\bar{\delta}[X]$ denotes a structural rank of matrix X defined as max rank $[X]$ over the set of all nonzero entries of X . The fixed mode of type 2 can be controllable under decentralized information structures if signalling from station α to station β is possible (assuming that (C, A, B) is both structurally controllable and observable).

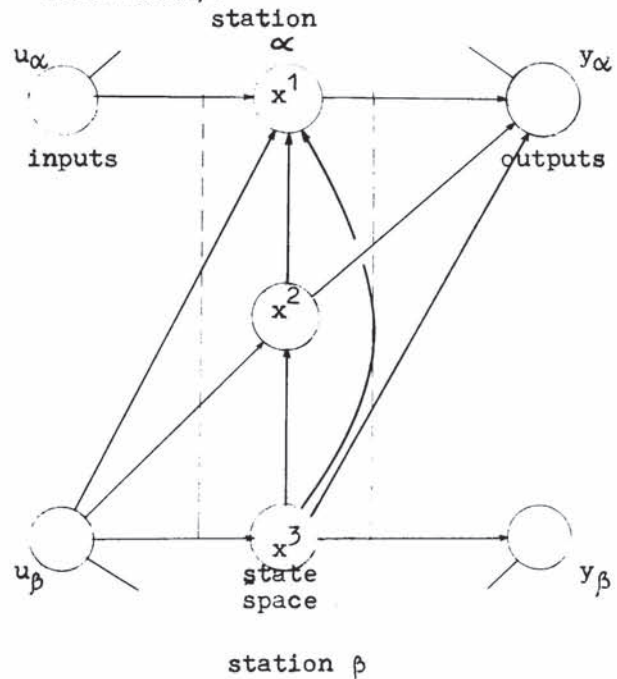


Fig.2. Accessibility diagram of system of fig.1.

The modes associated with the second group of state variables x^2 are structurally fixed modes of fig.1 - uncontrollable under the decentralized information structure. The arrow between the group of variables in this diagram shows that there are some non-zero entries in the matrix linking these groups of variables.

Having understood better the nature of fixed modes, we can now consider the following situations:

1) There are no fixed modes. In this case, according to theorem 1, pole placement under controller structure (2) is possible and thus we can use such decentralized linear time-invariant controllers. This is also the case when there exist fixed modes but which are sufficiently stable.

2) There are fixed unstable modes (or fixed stable modes but very close to the imaginary axis) which at the same time are controllable under the decentralized information structure. In this case there are two options. One of these is to design decentralized control laws (nonlinear and/or nonstationary) under which the system (1) is stabilised. This can be done by using explicit signalling strategies, Kobayashi et al [1978], or - which seems to be more practical - by applying periodic time-varying feedback laws, Anderson and Moore [1981] Purviance and Tyler [1982], or finally by putting a sampler-decay holder in series with each input of the system and then by stabilizing the resulting discrete time system, Wang [1982]. In particular, consider the result of Anderson and Moore [1981] for the system (1) with two control stations

($v=2$) and assume that $([C_1^T C_2^T]^T, A, [B_1, B_2])$ is controllable and observable.

Then, even if the fixed modes exist, the system can be made controllable through, say, u_1 and observable through y_1 by using time varying output feedback control law

$$u_2(t) = K_2(t)y_2(t),$$

where $K_2(t)$ is a piece-wise-constant periodic function with the number of segments $l_i > 1 + \max\{m_i, r_i\}$ if and only if $C_i[sI - A]^{-1}B_j \neq 0$ for $i, j=1, 2$ $i \neq j$ (compare this condition with (6)).

The second option exists in the case when the information structure can be changed so that appropriate data could be exchanged between the control stations through information channels. It is obvious that we would like to exchange as little information as possible. In terms of eliminating the fixed modes, we can represent this transfer of information by adding certain non-zero elements (off block-diagonal) in the feedback matrix K (see (3)). In other words, we can consider a set $\tilde{K}$ of output feedback matrices such that $K \in \tilde{K}$ and that there are no fixed modes of system (1) with

respect to output feedback matrices belonging to $\tilde{K}$. Armentano and Singh [1982] suggested a constructive procedure for eliminating the unstable fixed modes by introducing a sufficient number of informational links between the control stations.

3) If there are unstable fixed modes which are uncontrollable under the decentralized information structure (for example structurally fixed modes of type 1) then the exchange of information between control stations remains the only option available which allows us to stabilize the system.

In view of the above discussion and presentation of basic results concerned with the notion of the fixed modes, it would appear that

(i) One should avoid having fixed modes (and especially structurally fixed modes) when decomposing system (1) - in particular each eigenvalue of A should be controllable and observable from some control station.

(ii) If there are fixed modes then one should introduce appropriate information links between control stations in order to be able to move these modes - since by time varying feedback we can only get around the effects of those fixed modes which are controllable under the decentralized information structure and, in any case, it is not known how much feedbacks will be effected by disturbances acting upon a system.

REFERENCE BASED DESIGN TECHNIQUES FOR DETERMINISTIC DECENTRALIZED CONTROL

We saw in the last section that efforts to stabilize systems using decentralized linear invariant time feedback controllers can sometimes lead to unstable fixed modes. Here we will consider the case when this does not occur and discuss some approaches for the deterministic controller design. First we consider the approach outlined by Bernussou and Titli [1982] which is similar to a number of other design techniques based upon parametric optimization. Consider the following linear interconnected system(1)

$$\dot{x}_i(t) = \sum_{j=1}^v A_{ij}x_j(t) + B_{ii}u_i(t), \quad x_i(0) = x_{i0} \quad (8a)$$

$$y_i(t) = C_{ii}x_i(t) \quad (8b)$$

$$i=1, \dots, v$$

where $x_i \in R^{n_i}, u_i \in R^{r_i}, y_i \in R^{m_i}$ and

let $n = n_1 + \dots + n_v$. We want to find a linear static output decentralized feedback controller of the form

$$u_i = -K_i y_i, \quad i=1, \dots, v \quad (9)$$

such that the quadratic criterion (chosen initially in an additively separable form)

$$J = \sum_{i=1}^v \int_0^T (x_i^T Q_i x_i + u_i^T R_i u_i) dt + x_i^T(T) S_i x_i(T) \quad (10)$$

is minimized.

Along with information constraints as imposed by (9) we may want the local feedback matrices K_i to satisfy some additional linear constraints, Bernussou and Titli [1982]. These will not be considered here for the sake of simplicity. In most typical design cases, we would also make T go to infinity and in order to get rid of the influence of the initial conditions x_{i0} on the optimal decentralized gains - to assume that $x(0)$ is distributed uniformly over an n -dimensional sphere. Then the design problem can be transformed to the following form:

$$\min \hat{J}(K) = \text{tr } P \quad (11a)$$

$$K \in K_S$$

subject to

$$(A-BKC)^T P + P(A-BKC) + Q + C^T K^T R K C = 0 \quad (11b)$$

where $K_S \subset \mathbb{K}$ is the class of block diagonal output feedback matrices (see (3)) for which $(A-BKC)$ is a matrix with eigenvalues in the left hand plane and where A is composed of blocks A_{ij} , B = block diagonal $[B_{11}, \dots, B_{vv}]$, C = block-diagonal $[C_{11}, \dots, C_{vv}]$, Q = block-diagonal $[Q_1, \dots, Q_v]$, R = block-diagonal $[R_1, \dots, R_v]$. Thus this approach is applicable in the case when the class K_S is nonempty. This issue has been studied by many authors and the nonemptiness property can be checked for example by using a suitable initialisation algorithm (see for example Armentano and Singh [1981]; (stage 1 of the design procedure) in order to find some $K^0 \in K_S$ and we may compute the gradient of $\hat{J}(K)$ with respect to K at K^1 :

$$\frac{d\hat{J}(K^1)}{dK} = 2[RKC - B^T P]VC^T, \quad (12)$$

where P is a solution of eq.(11b) and V is a solution of a Lyapunov

equation.

$$(A-BKC)V + V(A-BKC)^T + I = 0$$

Projection of the matrix $d\hat{J}(K^1)/dK$ on the sub-space of block-diagonal matrices consists simply of taking the diagonal blocks of this matrix and it specifies the "matrix search direction"

$$D^1 = \text{block-diagonal } [d\hat{J}(K^1)/dK]$$

Then the choice of a step α^1 of progression needs to be made so as to get $K^{1+1} = K^1 - \alpha^1 D^1$ such that

$$\hat{J}(K^{1+1}) < \hat{J}(K^1) \quad \text{and} \quad K^{1+1} \in K_S$$

and the iteration is continued until an appropriate test for convergence is satisfied for example $\|D^1\| < \epsilon$. More details and examples of applications of the above procedure are given by Bernussou and Titli [1982].

A number of other approaches have also been developed for the synthesis of decentralized control through, for example:

- (a) hierarchical optimization and decentralized control with a pre-specified degree of stability, Hassan et al [1979],
- (b) decentralized control using model followers, Hassan and Singh [1980],
- (c) overlapping decompositions (overlapping information sets), Siljak [1979], Ikeda et al [1981].

All these methods were described in Chapter 6 of the book of Singh [1981]. We will briefly discuss the approach (b).

The eq. (8a) describing the interconnected system can be transformed into the following form:

$$\dot{x}_i = A_{ii}x_i + B_i u_i + G_i z_i, \quad i=1, \dots, v \quad (13a)$$

$$z_i = \sum_{j=1}^v L_{ij} x_j \quad (13b)$$

It is assumed that $y_i = x_i$, $i=1, \dots, v$. The approach requires the introduction of a crude "interaction model" of the form

$$\dot{z}_i = A_{zi} z_i \quad (14)$$

In Hassan and Singh, [1980] a way of choosing A_{zi} is suggested.

The structure of the local dynamic controller is shown in Fig.3, where gains K_i^1 , K_i^2 are chosen by solving appropriate optimisation problems, Hassan and Singh [1980]. The control

operating point or, finally, when the deterministic design results with ... an optimal solution of, for example, a stochastic control problem. Let us consider now this stochastic control problem within a decentralized control design framework. Since a rigorous description of a continuous time stochastic control problem requires special notation and since most of the useful results in decentralized stochastic control concern the discrete time case, we will also consider such a case here.

Thus we consider the linear invariant discrete time system being controlled from v control stations

$$x(k+1) = Ax(k) + \sum_{i=1}^v B_i u_i(k) + \xi(k) \quad (15a)$$

$$w(k) = Cx(k), \quad k=0, 1, \dots, N-1 \quad (15b)$$

where (15a) is the state equation, (15b) is the output equation, $x(k) \in R^n$, $u_i(k) \in R^{r_i}$, $w(k) \in R^m$. Each control station has access to the measurements

$$y_i(k) = H_i w(k) + \theta_i(k), \quad (16)$$

where

$$y_i(k) \in R^{m_i}.$$

$\xi(k)$, $\theta_i(k)$, $i=1, \dots, v$ are assumed to be white noise processes independent of the initial state $x(0)$ which is assumed to be a random vector. The classical information pattern arises when

$$H_i = I \text{ and } \theta_i(k) = \theta(k) \text{ for } i=1, \dots, v \quad (17)$$

where I is the identity matrix. This implies that all the controllers have identical on-line information.

A non-classical decentralized information pattern arises when the above condition (17) is not satisfied; in particular, when the controls are restricted to be of the form

$$u_i(k) = \gamma_i^k(y_i^k),$$

where

$$y_i^k = (y_i(0), \dots, y_i(k)) \quad (18)$$

i.e. the controls at the i th control station are chosen as some function γ_i^k of the past and present information available to this control station.

Let us also consider the quadratic performance

$$J = E \left[\sum_{k=0}^{N-1} (x^T(k) Q_k x(k) + \sum_{i=1}^v u_i^T(k) R_i u_i(k)) + x^T(N) S x(N) \right], \quad (19)$$

where Q_k, S are symmetric positive semi-definite matrices and $R = \text{block-diagonal } [R_1, \dots, R_v]$ is a symmetric positive definite matrix. The control rules (18) should be chosen so as to minimize performance index (19).

Witsenhausen [1968] considered a seemingly simple example to show that for non-classical information patterns, the optimal control law is not linear and that the optimal estimator may be infinite dimensional. The reason for this is that due to the lack of possibility of directly transferring information between the controllers so as to reduce the overall cost, the information transfer is attempted through the dynamics of the system. The optimal control law requires each controller to use its control actions, not only to exercise control, but also to convey information to the other controllers. The interaction between the control and the signalling effect is necessarily non-linear!

A lot of work has been done in the last several years to seek out special cases where linear solutions are possible and/or where the separation between state estimation and computing control inputs can be achieved: Ho and Chu [1974, 1972], Ho [1980] found that the so-called "partially nested information structures" lead to decision rules which are linear. In broad terms, the partially nested information structure can be defined as follows. If control action $u_i(k)$ affects the observation $y_j(1)$ then the j th control station at time 1 should know what the i th control station knows at time k . The reason why linear decision rules are optimal in this case is because no signalling is necessary. An important example of a problem with partially nested information structure from the point of view of control applications was investigated by Kurtaran and Sivan [1974]. They considered a one-step delay information sharing pattern, where the local control rules can be of the form

$$u_i(k) = \gamma_i^{k-1}(y_1^{k-1}, \dots, y_{i-1}^{k-1}, y_i^k, y_{i+1}^{k-1}, \dots, y_v^{k-1}) \quad (20)$$

Thus a one-step delay sharing pattern means that each controller has all its observations up to the current time step y_i^k and all the previous observations (y_j^{k-1} , $j=1, \dots, v$ $j \neq i$) of the other controllers. This also means that each controller knows the previous control actions of the other controllers - either they are transmitted to it or their values can be recovered using the system model and measurement data. Therefore, the control law (20) can also be written down as

$$u_i(k) = \gamma_i^k(y_i(k), \delta^{k-1}), \quad (21)$$

where

$$\delta^{k-1} = (y_1^{k-1}, \dots, y_v^{k-1}, u_1^{k-1}, \dots, u_v^{k-1}) \quad (22)$$

1

$(\delta^{-1} = \phi)$.

The main result of Kurtaran and Sivan is:

Theorem 3

For the one-step delay information sharing pattern the optimal controller is given by

$$u_i(k) = \hat{F}_i^k y_i(k) + \hat{G}_i^k x(k|k-1), \quad i=1, \dots, v \quad (23)$$

where $\hat{F}_i^k, \hat{G}_i^k$ are gain matrices and $x(k|k-1)$ is the conditional expectation of $x(k)$ given δ^{k-1} .

The optimal sequences of gain matrices $\hat{F}_i^k, \hat{G}_i^k$ as well as filter gains used in order to compute $x(k|k-1)$ can be computed from recursive relations (Singh [1981]). We should note that the separation property implied by (22) is different from the classical certainty equivalence property.

In particular, $\hat{F}_i^k, \hat{G}_i^k$ cannot be computed through a deterministic design approach (assuming that $x(k-1)$ is known at time k for all control stations). We should also note that each control station needs to use on-line the full system model in order to compute the estimates of the overall state vector $x(k)$. Yet the computing of $x(k|k-1)$ can be done between the times $k-1$ and k .

The results concerning the separation between the estimation and control for decentralized stochastic control systems were obtained also for information structures having the so-called sequential partitions (s-partitions), Yoshikawa and Hirodki [1978]. Without going into details, we may

say in particular that the information structure has an s-partition if the local control laws are constrained to be of the form:

$$u_i(k) = \gamma_i^k(y_i(1), u_i(1-1); y(p), u(p-1); \\ ; k_m < 1 < k, \dots, 0 < p < k_m). \quad (24)$$

where $k_m < k < k_{m+1}$. This means that all previous information is exchanged between the controllers at times k_m . Application of the result obtained by Yoshikawa and Hirodki [1978] to stochastic control problem as given by eqs. (15,16,19) yields the following separation theorem:

Theorem 4

The optimal control laws under s-partition information sharing pattern have the form ($k_m < k < k_{m+1}$)

$$u_i(k) = \gamma_i^k(y_i(1), u_i(1-1), q_m); \quad k_m < 1 < k_m \quad (25)$$

where $q_m \stackrel{\text{def}}{=} p(x(k_m+1) | y(p), u(p-1); 0 < p < k_m)$ is the conditional probability density of $x(k_m+1)$ given all measurement data and controls up to time k_m .

Thus the past data can be reduced to q_m . This advantage becomes especially important when q_m has an equivalent statistic q_m , Streibel [1965]. It should be noted that control laws (25) are nonlinear in general. When the above result is specialised to one-step delay information sharing pattern, then the control laws become linear (Thm.3) - since in such a case no signalling is necessary. At the same time the costs associated with exchange of information between the controllers are increased.

Unfortunately there are no results available concerning the properties of optimal control law under decentralized information structures (without any exchange of on-line information between the control stations). In practical design problems, we have to use suboptimal techniques: enforce separation between state estimation and computing of the control inputs and to specify some fixed structure (in most cases linear) of the control laws

Such an approach was adopted by Chong and Athans [1971] who constrained the controls to be linear functions of local estimates of the full state vector obtained through local full-dimensional filters being driven by lo-

cal measurements. The filter gain matrices and control feedback matrices need to be computed together centrally by solving a matrix two-point boundary value problem of large dimension. The local filters are not the optimal Kalmanfilters although they are linear and unbiased. It also appears that we are paying for the decentralized information pattern by a considerable increase in the off-line computations and by running on-line v full-dimensional filters.

A further step of simplification is to enforce the certainty equivalence between state estimation and control design. Using such an approach, we can use deterministic design techniques (see section 4 of this paper). Then we need to develop appropriate decentralized state estimators (filters). In order to get unbiased estimates of local state variables, we have the following possibilities:

- (1) to build full-dimensional local filters driven by local information as in the approach of Chong and Athans [1971].
- (2) to build local filters of lower dimension - yielding the estimates of local states only. In this case, however, it is necessary to exchange information between local filters (see Singh [1981]) for a description of these techniques).
- (3) to measure interaction input for the subsystems and then to run local low dimensional filters driven by local measurements (including the interaction measurements). This approach was developed by Tucker, Sanders and Linton in a series of papers (see for example, Sanders et al [1978]). A certain amount of information exchange between local filters may improve their performance but it may also increase the costs of operation.

Finally, it should be mentioned that some interesting work has also been done by Athans and his co-workers for the case when the decentralized control laws are periodically updated by the coordinator unit, Chong and Athans [1978]. We will not discuss this approach here because we are concerned in this paper with pure decentralized control with no coordinator unit and because the control scheme as developed by Chong and Athans requires rather extensive transfer of information between the coordinator and the local control stations

The stochastic control problem does not represent the only way of defining control design problems under uncertainty. In numerous applications, it will be much more suitable to con-

sider the so-called wave-form disturbances which can be described as outputs of a dynamical system with unknown initial conditions and with occasional jumps in the values of state variables. In a series of papers on the decentralized servomechanism problem, Davison and his co-workers have adopted such an approach. Because of lack of space, we will not present here this approach in more detail. We shall only note that it consists of designing decentralized servo-compensators in order to eliminate the offset errors due to persistent disturbances and then of designing stabilizing compensators by decentralized design techniques. On-line tuning of the local controllers is possible and has been investigated in a series of papers (see for example Davison [1978]). The results concerning the above approach are summarized in a recent paper, Davison [1982] dealing with the decentralized robust servomechanism problem using local models.

Finally, one may also consider problems with set bounded disturbances and there is a need to develop more techniques for handling such problems in a decentralized framework.

ROBUSTNESS ISSUES AND FINAL CONCLUSIONS

The majority of existing results and design techniques for decentralized control do not address explicitly the reliability issue. From a practical point of view, it is well known that the control systems should possess various robustness properties. In classical design procedures for single-input-single-output servomechanisms, the well known concepts of gain and phase margins are used to provide for sufficiently reliable controller operation. It is therefore obvious that after establishing a variety of techniques for decentralized control design, there is now a considerable interest in providing for the designed controller to be reliable. First of all, it is necessary to specify what we mean by controller reliability and by its robustness properties. Let us consider briefly some possible sources of failure in the operation of the control structure.

- (1) The plant model as used for a design technique may not and usually will not describe exactly the behaviour of the plant.
- (2) The plant behaviour can change during its operation. These changes may be either slow in time (for example, a catalyst activity can decay during chemical plant operation)

or may have an abrupt character when some failure occurs.

- 3) The sensors and the actuators can fail to work properly on certain occasions.
- 4) The information links can be broken.
- 5) The controller itself may suffer some degree of failure.

The above statements are rather trivial but when put together, they allow us to understand that it would have been hopeless in general to aim at designing control structures which could withstand all possible failures. Therefore the robustness of a given design has to be defined with respect to a certain controller property or properties which should be preserved under specified classes of perturbations and/or failures. For example, Davison [1976] when considering the robust decentralized controller design for a general servomechanism problem, states the robustness property as follows: ... Under process parameter variations the controller will regulate the outputs to their reference values as long as the system remains stable. Siljak [1980] and in a series of more recent papers by himself and his co-workers, considers robustness of a large-scale system control structure with respect to stability and suboptimality under structural perturbations defined as disruptions of interconnections between the subsystems. He also considers the suboptimal design techniques which could lead to control structures being immune to certain distortions of the local control laws (see, for example, Siljak [1980]) and which provide for certain gain and phase margins. More examples of addressing the robustness of design can be given but we do not aim here to provide for a complete list of these. In our opinion, there is a limited possibility of incorporating all important robustness issues into one systematic design method defined in mathematical terms. It is definitely worthwhile and necessary to introduce control algorithms which could cope with changing - within some prescribed region - plant parameters. In fact, some results concerning the control of systems with randomly jumping parameters can already be reported. The ideas of self-tuning controllers can perhaps be used and developed within the decentralized control framework. Yet, one should be aware of the fact that by increasing robustness (in an appropriate sense) of a control law we will usually obtain a controller which performs less efficiently under normal (non-failure) conditions. There-

fore it seems to us that in practical applications, we should think rather in terms of protecting the control structure against perturbations and failures by developing and implementing the jacketing hardware and software. This could include failure detection circuitry and algorithms followed by procedures for reconfiguration of the control structure after a perturbation requiring such reconfiguration occurs and is detected. Standby redundant controllers can be used to protect the system against the failure of operating controllers. To conclude the above remarks, it should be noted that undoubtedly the reliability issue is one of the most important future research directions in the decentralized control area.

Let us finally summarize, in a few points, our conclusions regarding the decentralized control problem:

- i) There are numerous design techniques available; however one can only cope with linear time-invariant problems. We feel that future research should be oriented towards developing decentralized design methods which may be used to solve control problems belonging to a certain area of applications. In other terms, one can hope to achieve more constructive results related, for example, to decentralized control of multireservoir water systems or to nonlinear servomechanism problems while taking into account the specific features of a given field of applications than by continuing with broad theoretical studies.
- (ii) More research is needed as far as defining and investigating robustness properties of design techniques and control algorithms are concerned. In particular, it is worthwhile to exploit the idea of self-tuning controllers in a decentralized control framework.
- (iii) An appropriate approach to designing the jacketing hardware and software is required in order to provide for reliability of decentralized control structures for large-scale systems.
- (iv) The benefit-versus-cost analysis is required with respect to the value of exchanging information between the control stations.

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